WORKED SOLUTION

STEP III 2022, Question 11

Probability and Statistics • Binomial Mean Deviation

STEP III 2022, Question 11

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

A fair coin makes the distribution of X symmetric about its mean, so the probabilities below the mean match those above it. That symmetry turns the absolute value into a one-sided sum. The identity in (i)(c) then turns $\sum r\binom{2n}{r}$ into a tail sum of a smaller binomial distribution, which symmetry evaluates again.

Solution

Throughout, $X \sim \mathrm{B}(N, \frac{1}{2})$, so $\mathrm{P}(X = r) = \binom{N}{r}2^{-N}$ and $\mu = \frac{1}{2}N$. Because $\binom{N}{r} = \binom{N}{N-r}$, we have $\mathrm{P}(X = r) = \mathrm{P}(X = N - r)$: the distribution is symmetric about $\frac{1}{2}N$.

Part (i)(a)

With $N = 2n$, the map $r \mapsto 2n - r$ sends $\{0, \ldots, n-1\}$ to $\{n+1, \ldots, 2n\}$, so by symmetry

$$\mathrm{P}(X \leq n-1) = \mathrm{P}(X \geq n+1).$$

The three events $X \leq n-1$, $X = n$, $X \geq n+1$ cover every outcome, so $2\,\mathrm{P}(X \leq n-1) + \mathrm{P}(X = n) = 1$, which gives

$$\mathrm{P}(X \leq n-1) = \tfrac{1}{2}(1 - \mathrm{P}(X = n)).$$

Part (i)(b)

Here $\mu = n$, so

$$\delta = \sum_{r=0}^{2n} |r - n| \binom{2n}{r} \frac{1}{2^{2n}}.$$

The term $r = n$ is zero. Substituting $r = 2n - s$ in the terms with $r \geq n+1$ gives $|r - n| = n - s$ and $\binom{2n}{r} = \binom{2n}{s}$ with $0 \leq s \leq n-1$. So the terms above the mean exactly

repeat those below it:

$$\delta = 2\sum_{r=0}^{n-1}(n-r)\binom{2n}{r}\frac{1}{2^{2n}} = \sum_{r=0}^{n-1}(n-r)\binom{2n}{r}\frac{1}{2^{2n-1}},$$

as required.

Part (i)(c)

For $1 \le r \le 2n$,

$$r\binom{2n}{r} = \frac{r\,(2n)!}{r!\,(2n-r)!} = 2n\cdot\frac{(2n-1)!}{(r-1)!\,(2n-r)!} = 2n\binom{2n-1}{r-1}.$$

Now split the sum in (b) as $2^{2n-1}\delta = A - B$, where

$$A = n\sum_{r=0}^{n-1}\binom{2n}{r}, \qquad B = \sum_{r=1}^{n-1} r\binom{2n}{r}$$

(the $r = 0$ term of B is zero, so we drop it).

The sum A. By (a), $\sum_{r=0}^{n-1}\binom{2n}{r} = 2^{2n}\,\mathrm{P}(X \le n-1) = 2^{2n-1} - \frac{1}{2}\binom{2n}{n}$, so

$$A = n\,2^{2n-1} - \tfrac{n}{2}\binom{2n}{n}.$$

Ask yourself How do we add up $\binom{2n-1}{s}$ for s from 0 to $n-2$?

The sum B. By the identity, with $s = r - 1$,

$$B = 2n\sum_{s=0}^{n-2}\binom{2n-1}{s}.$$

The row $\binom{2n-1}{s}$, $0 \le s \le 2n-1$, has an even number of entries and adds up to 2^{2n-1}. It is symmetric under $s \mapsto 2n-1-s$, which swaps $\{0,\dots,n-1\}$ with $\{n,\dots,2n-1\}$, so the first half adds up to 2^{2n-2}. Our sum stops one short of that half, so

$$\sum_{s=0}^{n-2}\binom{2n-1}{s} = 2^{2n-2} - \binom{2n-1}{n-1} = 2^{2n-2} - \tfrac{1}{2}\binom{2n}{n},$$

using $\binom{2n}{n} = \binom{2n-1}{n-1} + \binom{2n-1}{n} = 2\binom{2n-1}{n-1}$. Hence

$$B = n\,2^{2n-1} - n\binom{2n}{n}.$$

Putting it together.

$$2^{2n-1}\delta = A - B = \left(n - \frac{n}{2}\right)\binom{2n}{n} = \frac{n}{2}\binom{2n}{n}, \qquad \text{so} \qquad \delta = \frac{n}{2^{2n}}\binom{2n}{n}.$$

Part (ii)

Now $N = 2n+1$ and $\mu = n + \frac{1}{2}$, which is not a possible value of X. The symmetry $r \mapsto 2n+1-r$ swaps $\{0,\dots,n\}$ with $\{n+1,\dots,2n+1\}$ and preserves both $|r-\mu|$ and $\binom{2n+1}{r}$. So, as in (i)(b),

$$\delta = 2\sum_{r=0}^{n}\left(n+\tfrac{1}{2}-r\right)\binom{2n+1}{r}\frac{1}{2^{2n+1}} = \frac{1}{2^{2n+1}}\sum_{r=0}^{n}(2n+1-2r)\binom{2n+1}{r}.$$

- By the same symmetry, the first half of the row $\binom{2n+1}{r}$ adds up to half of 2^{2n+1}: $\sum_{r=0}^{n}\binom{2n+1}{r} = 2^{2n}$.
- By the identity $r\binom{2n+1}{r} = (2n+1)\binom{2n}{r-1}$ (proved exactly as in (i)(c)), and then (i)(a),

$$\sum_{r=1}^{n} r\binom{2n+1}{r} = (2n+1)\sum_{s=0}^{n-1}\binom{2n}{s} = (2n+1)\left(2^{2n-1} - \tfrac{1}{2}\binom{2n}{n}\right).$$

Therefore

$$\sum_{r=0}^{n}(2n+1-2r)\binom{2n+1}{r} = (2n+1)2^{2n} - (2n+1)\left(2^{2n} - \binom{2n}{n}\right) = (2n+1)\binom{2n}{n},$$

and

$$\boxed{\delta = \frac{2n+1}{2^{2n+1}}\binom{2n}{n}}$$

when $N = 2n+1$.

Checks

- $N = 3$ ($n = 1$): $|X - \frac{3}{2}|$ is $\frac{3}{2}$ with probability $\frac{1}{4}$ and $\frac{1}{2}$ with probability $\frac{3}{4}$, so $\delta = \frac{3}{8} + \frac{3}{8} = \frac{3}{4}$. The formula gives $\frac{3\cdot 2}{8} = \frac{3}{4}$. With $n = 0$ ($N = 1$) it gives $\frac{1}{2}$, which is also right.
- $N = 4$ ($n = 2$): (i)(c) gives $\frac{2\cdot 6}{16} = \frac{3}{4}$, the same as for $N = 3$. In fact δ for $N = 2n+1$ always equals δ for $N = 2n+2$: since $\binom{2n+2}{n+1} = \frac{2(2n+1)}{n+1}\binom{2n}{n}$, the even formula with $n+1$ in place of n reduces to the odd one.
- Both formulas agree exactly with the definition of δ for every $N \le 60$, checked by computer; for example $N = 10$ gives $\delta = \frac{315}{256} \approx 1.230$.

Common slips

- Forgetting the factor 2 from the symmetry, which is where 2^{2n-1} rather than 2^{2n} comes from in (i)(b).
- Getting the range of s wrong after the shift $s = r-1$: the sum stops at $n-2$, one term short of half the row, and that missing term $\binom{2n-1}{n-1}$ is what makes the answer non-zero.

- In (ii), treating μ as an integer: no term vanishes, and both halves of the row have $n+1$ terms.

Beyond STEP

Stirling's formula gives $\binom{2n}{n} \approx 4^n/\sqrt{\pi n}$, so $\delta \approx \sqrt{N/(2\pi)}$ for large N. Since $\sigma = \frac{1}{2}\sqrt{N}$, this is $\delta \approx \sigma\sqrt{2/\pi}$, exactly the mean absolute deviation of a normal distribution, as the central limit theorem predicts. Grimmett and Welsh, *Probability: An Introduction*, covers both the normal approximation to the binomial and the central limit theorem behind it.

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